

Lesson 21: Fourier Series & Frequency Analysis

Professor Dave Explains Differential Equations Master Series · Lesson 21

Video: Fourier Series and Transforms Part 1: Concepts in Frequency Analysis **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains Jean-Baptiste Joseph Fourier's revolutionary discovery: any periodic signal, no matter how jagged, discontinuous, or complex (like a square wave or an audio synthesizer tone), can be dismantled into an infinite sum of simple pure sine and cosine harmonics! In music, this is the overtone series; in engineering, it is frequency analysis. By exploiting the mathematical orthogonality of sines and cosines, we compute the exact harmonic weight of every frequency in the signal.

2. Core Theoretical Framework & Rigorous Formulations

1. The Trigonometric Fourier Series on $[-L, L]$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

2. Euler-Fourier Formulas

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx, \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

3. Symmetry Exploitation Rules

- Even Functions** ($f(-x) = f(x)$): Symmetric about y -axis. $b_n = 0$ identically! $a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$. (Fourier Cosine Series)
- Odd Functions** ($f(-x) = -f(x)$): Anti-symmetric about origin. $a_0 = 0$, $a_n = 0$ identically! $b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$. (Fourier Sine Series)

3. Fully Solved Whiteboard Problems

Engineering Classic: Fourier Expansion of a Square Wave

Problem Statement: Compute the Fourier series of the periodic square wave $f(x) = \begin{cases} -1, & -\pi < x < 0 \\ +1, & 0 < x < \pi \end{cases}$ with period $T = 2\pi$ ($L = \pi$).

Step 1: Check Symmetry

Observe $f(-x) = -f(x)$. The function is strictly **Odd**! Therefore, $a_0 = 0$ and $a_n = 0$ for all $n \geq 1$ without doing any integration!

Step 2: Calculate Sine Coefficients b_n

$$b_n = \frac{2}{\pi} \int_0^{\pi} (1) \sin(nx) dx = \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2}{n\pi} [1 - \cos(n\pi)].$$

Step 3: Evaluate Parity of $\cos(n\pi) = (-1)^n$

If n is even ($n = 2, 4, 6$): $1 - (-1)^n = 1 - 1 = 0 \implies b_n = 0$. If n is odd ($n = 1, 3, 5$): $1 - (-1)^n = 1 - (-1) = 2 \implies b_n = \frac{4}{n\pi}$.

Step 4: Assemble Trigonometric Series

$$f(x) = \frac{4}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \frac{1}{7} \sin 7x + \dots \right).$$

Step 5: Bonus Mathematical Jewel (Leibniz Formula for π)

Evaluate at $x = \frac{\pi}{2}$ where $f(\frac{\pi}{2}) = 1$: $1 = \frac{4}{\pi} (1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots) \implies \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots!$

⚠ Common Exam Pitfall & Trapdoor Avoidance

Always check function parity (even vs odd) before computing integrals. Calculating a_n by hand for an odd function wastes 10 minutes on an exam only to arrive at zero.

4. Key Takeaways & Exam Strategy Formula Card

 Core Takeaway & Exam Rule

Odd functions generate pure sine series (b_n); even functions generate pure cosine series (a_n).

- Video reference: [Fourier Series and Transforms Part 1: Concepts in Frequency Analysis](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.